Worksheet of Geometric Sums & Induction

Finite and Infinite Geometric Sums

1. Evaluate the related sum of each sequence

(a) 2, 12, 72, 432.

(b) -1, 5, -25, 125.

(c) −2, 6, −18, 54, −162.

(d) −2, −12, −72, −432, −2592.

2. Evaluate

$$\sum_{k=0}^{5} -\left(-\frac{1}{3}\right)^{k}$$

3.

State whether the infinite series converges or diverges.

$$125 - 75 + 45 - 27 + \frac{81}{5} - \frac{243}{25} + \ldots$$

4.

Find the sum of the infinite series.

$$4 + \frac{16}{5} + \frac{64}{25} + \frac{256}{125} + \frac{1024}{625} + \ldots$$

5.

State whether the infinite series converges or diverges.

$$\sum_{n=1}^{\infty} 12\left(\frac{1}{2}\right)^{n}$$

6.

Choose all of the following series that diverge.

○ $\sum_{n=1}^{\infty} 10\left(-\frac{7}{5}\right)^n$ ○ $\sum_{n=1}^{\infty} (5)^n$ ○ $\sum_{n=1}^{\infty} 50\left(\frac{6}{5}\right)^n$

○ $\sum_{n=1}^{\infty} 4n-7$

7.

A company is offering a new subscription service. For promotional purposes, the company decides to offer the service at a discounted rate for the first year. The cost of the subscription starts at $\$84.00$ in the 1st month, $\$71.40$ in the 2nd month, $\$60.69$ in the 3rd month, and so on. Each month the cost is 85% of the previous month's cost. The promotion lasts 10 months. How much will a customer pay in total?
The total cost can be found using the formula for the sum of a finite geometric series:

$$S_n = \frac{a_1\left(1-r^n\right)}{1-r}$$

Round your answer to the nearest whole number.

8. Determine if each infinite geometric series converges or diverges. If converges, then find sum.

1) $a_1 = -3,\ r = 4$

2) $a_1 = 4,\ r = -\frac{3}{4}$

3) $a_1 = 5.5,\ r = 0.5$

4) $a_1 = -1,\ r = 3$

5) $81 + 27 + 9 + 3...,$

6) $7.1 + 17.75 + 44.375 + 110.9375...,$

9. Evaluate

$$\sum_{k=1}^{\infty} \frac{16}{9}\left(\frac{3}{2}\right)^{k-1}$$

10. Evaluate

$$\sum_{k=1}^{\infty} \frac{7}{6}\left(-\frac{1}{4}\right)^{k-1}$$

Induction

1.

Sums of Geometric Progressions Use mathematical induction to prove this formula for the sum of a finite number of terms of a geometric progression with initial term a and common ratio r:

$$\sum_{j=0}^{n} ar^j = a + ar + ar^2 + \cdots + ar^n = \frac{ar^{n+1} - a}{r - 1} \qquad \text{when } r \neq 1,$$

where n is a nonnegative integer.

2.

Use mathematical induction to prove the inequality

$$n < 2^n$$

for all positive integers n.

3.

Use mathematical induction to prove that $2^n < n!$ for every integer n with $n \geq 4$. (Note that this inequality is false for $n = 1, 2$, and 3.)

4.

An Inequality for Harmonic Numbers The **harmonic numbers** H_j, $j = 1, 2, 3, \ldots$, are defined by

$$H_j = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{j}.$$

For instance,

$$H_4 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{25}{12}.$$

Use mathematical induction to show that

$$H_{2^n} \geq 1 + \frac{n}{2},$$

whenever n is a nonnegative integer.

5.

Use mathematical induction to prove that $n^3 - n$ is divisible by 3 whenever n is a positive integer.

6.

Use mathematical induction to prove that $7^{n+2} + 8^{2n+1}$ is divisible by 57 for every nonnegative integer n.

7.

Use mathematical induction to prove the following generalization of one of De Morgan's laws:

$$\overline{\bigcap_{j=1}^{n} A_j} = \bigcup_{j=1}^{n} \overline{A_j}$$

whenever $A_1, A_2, \ldots, A_n$ are subsets of a universal set U and $n \geq 2$.

8.

Prove that $1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n+1)! - 1$ whenever n is a positive integer.

9.

Prove that for every positive integer n,

$$1 \cdot 2 + 2 \cdot 3 + \cdots + n(n+1) = n(n+1)(n+2)/3.$$

10.

Prove that if $A_1, A_2, \ldots, A_n$ and $B_1, B_2, \ldots, B_n$ are sets such that $A_j \subseteq B_j$ for $j = 1, 2, \ldots, n$, then

$$\bigcup_{j=1}^{n} A_j \subseteq \bigcup_{j=1}^{n} B_j.$$

11.

Prove that if $A_1, A_2, \ldots, A_n$ and $B_1, B_2, \ldots, B_n$ are sets such that $A_j \subseteq B_j$ for $j = 1, 2, \ldots, n$, then

$$\bigcap_{j=1}^{n} A_j \subseteq \bigcap_{j=1}^{n} B_j.$$

12.

ICP 10-6 Prove that

Theorem

The sum of cubes of first n positive integers is $\left(\frac{n(n+1)}{2}\right)^2$

13.

ICP 10-7 Prove that

Theorem

An ATM with only \$3 and \$7 bills can generate any amount $n \geq 12$

Basis Step: $n = 12$ dollar. $\triangleright$ $4 \times \$3$

Induction Step: Prove if ATM can generate $\$n$, then it can generate $\$n + 1$

Case 1: Output of $\$n$ contains $\geq$ two \$3 bills

Case 2: Output of $\$n$ contains $\leq$ one \$3 bill, it must have $\geq$ two \$7 bills